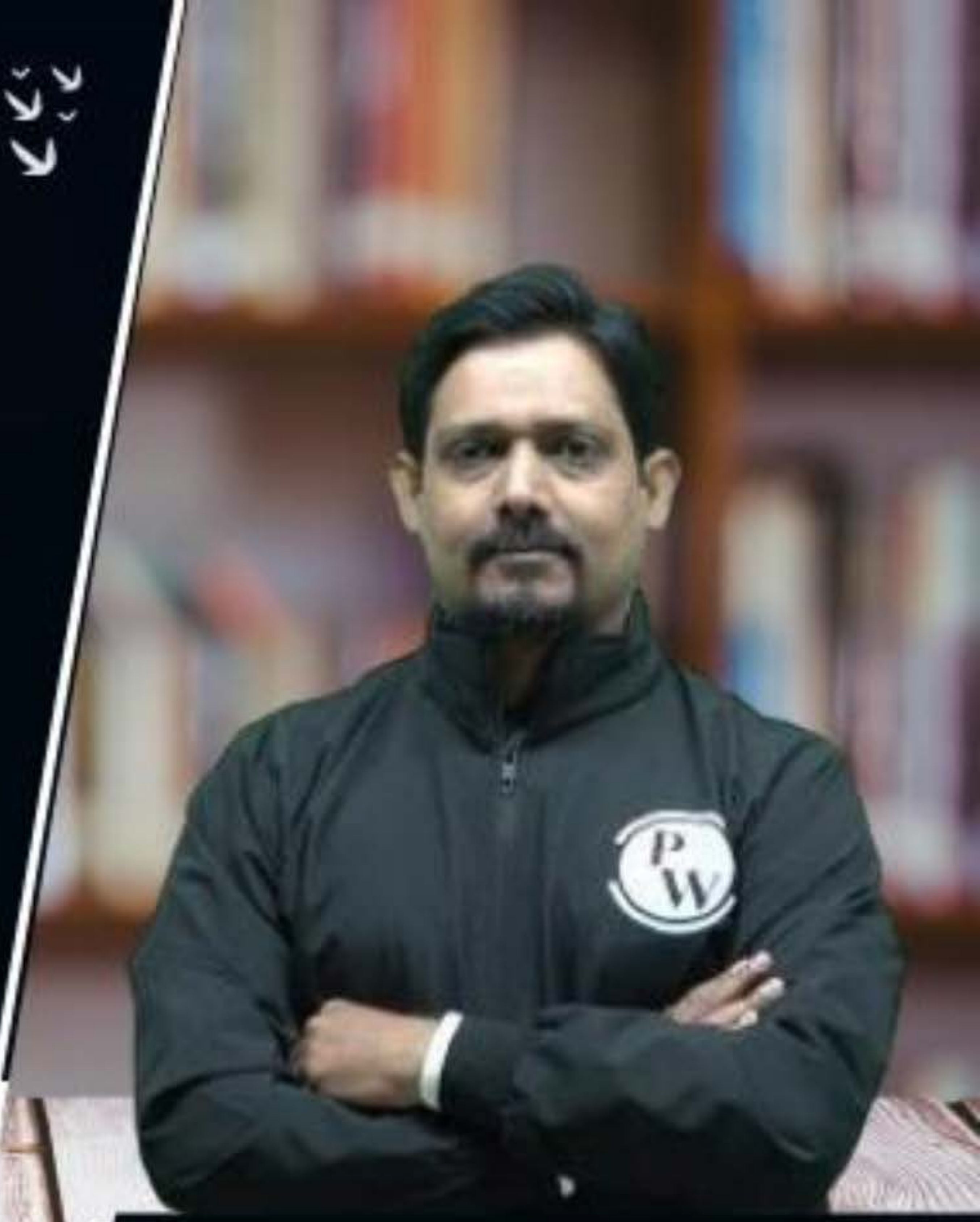
Computer Science & IT

Data Structure & Programming

Graph & Hashing

Lecture No. 03



By- Abhishek Sir

Recap of Previous Lecture

Topic Quadratic probing

Topic Double Hashing

Topic

Topic

Topic

Slide

Topics to be Covered

Topic — Graph Representation

Topic — Breadth first search

Topic

Topic

Topic

Slide

Topic : Graph

G is set of vertex and edge $G(V,E)$

problem where Graph is used as model

Hamitonian cycle

1. Connectivity
2. Networking
3. Shortest path
4. Minimimum

4. Register allocation
   Graph coloring
5. Chemistery :- two compound same formula but moleculor structure
   Isomer : Isomorphism

Slide

Topic : Graph

Simple Graph : No self loop No multiple edges

Graph Representation

1. Adjacency List
2. Adjacency Matrix

Adjacency ??

Direct edge to vertex is called Adjacency List

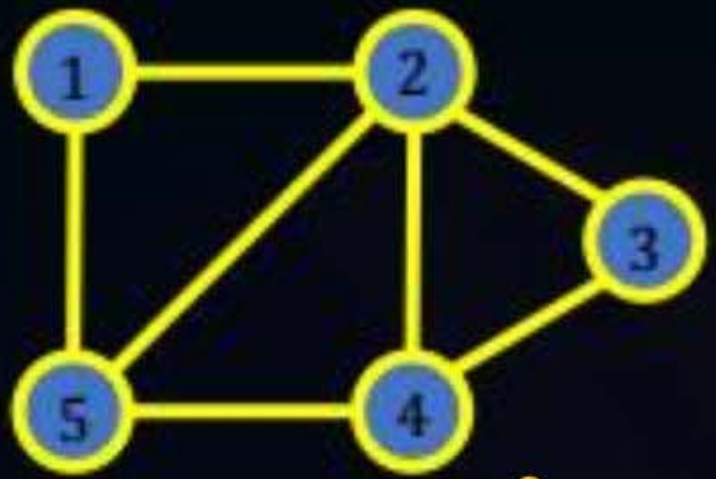
Slide

Topic : Graph

Adjacency List

traversal complexity : $\Theta(V+E)$

Last year PYQ order was defined



1 → 2 — 5

2 → 1 → 5 → 4 → 3

3 → 2 — 4

4 → 2 → 3 → 5

5 → 1 → 2 → 4

Length of List = degree of a Node

$\Theta(V+2E)$

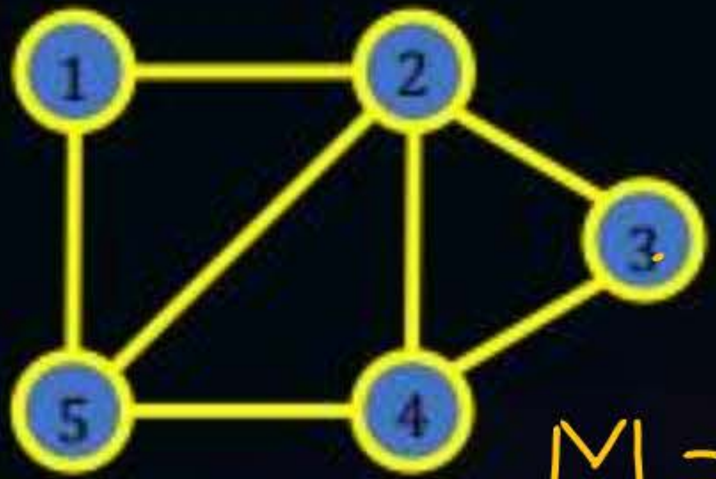
$\Theta(V+E)$

Space Complexity

$\Theta($No of HP + Length of List$)$

$\Theta\left(V + \sum_{i=1}^{V} deg(v_i)\right)$

Topic : Adjacency Matrix



$$M = \begin{array}{c|ccccc} & 1 & 2 & 3 & 4 & 5 \\ \hline 1 & 0 & 1 & 0 & 0 & 1 \\ 2 & 1 & 0 & 1 & 1 & 1 \\ 3 & 0 & 1 & 0 & 1 & 0 \\ 4 & 0 & 1 & 1 & 0 & 1 \\ 5 & 1 & 1 & 0 & 1 & 0 \end{array}$$

$M[i,j] = 1$

if $(i,j) \in E$

otherwise

$M[i,j] = 0$

M is tranpose Matrix

$M = M^T$

Space complexity $\Theta(|V|^2)$

Topic : Graph

Graph traversal : Traversal is process of visiting each node of Graph

1. Breadth first Search
2. Depth first Search

BFS : After visiting a node Adjacent node of vertex will be visited and repeat the process

Slide

Topic : Graph

We will maintain
An Array called
visited

All assigned
to 0

1 start visiting Source vertex

1

mark it as visited

Add Adjacent Node

1	1
2	3

to Queue.

We also maintain
the information of
reachability of a vertex
through another vertex

Slide

Topic : Graph

We will maintain An Array called visited

All assigned to 0

1	1
2	3

3	

1	2	2
3	4	5

delete 3

2	2	
4	5	

2	2	3	3
4	5	6	7

Add the Adjacent node

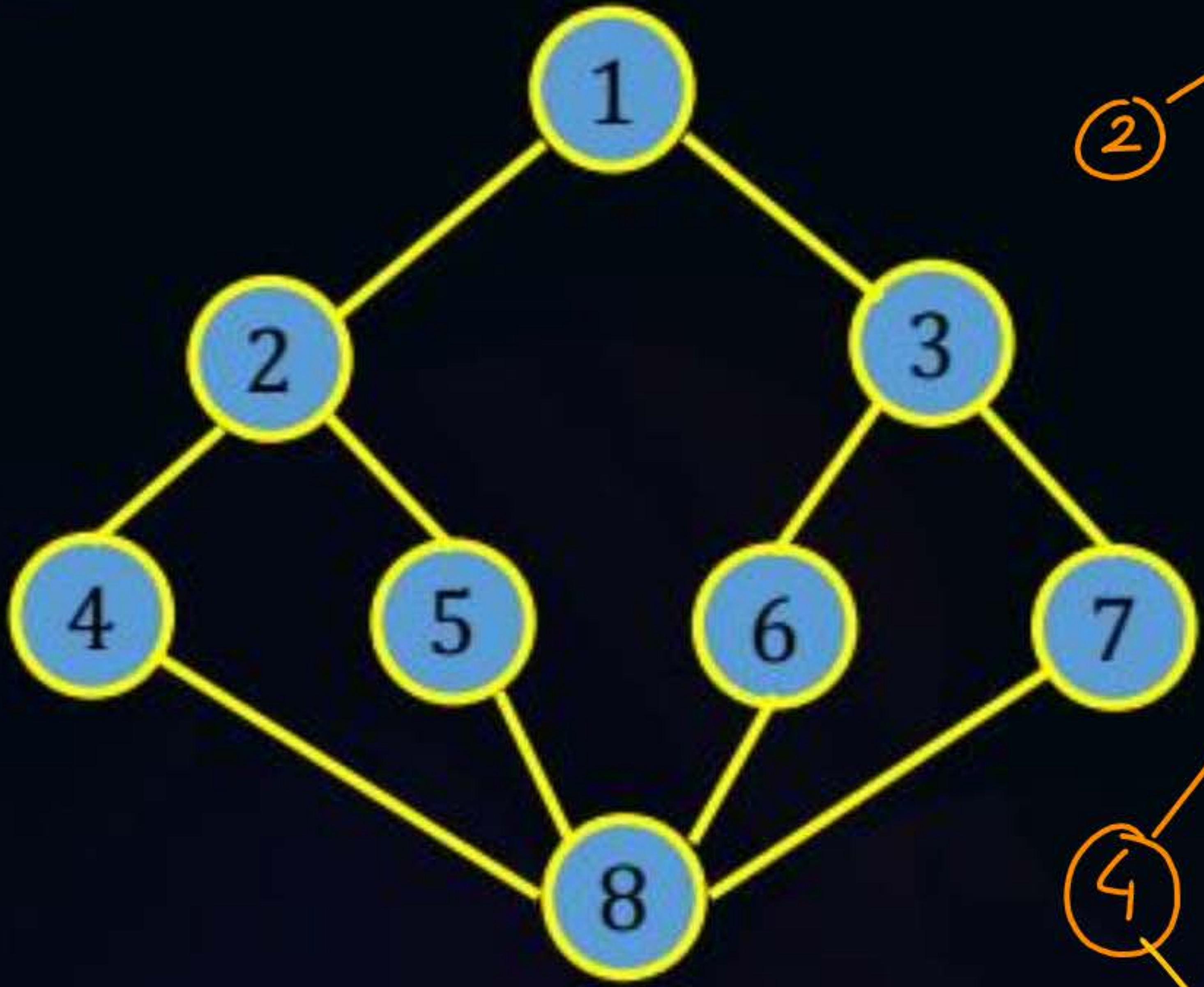
if Queue is empty BFS finishes

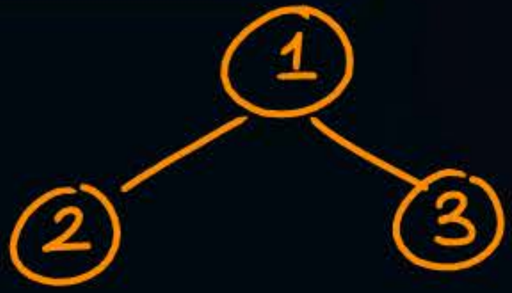
otherwise delete an element from Queue and repeat the process

(Add the Adjacent Node)

Slide

Topic : Graph





2	2	3	3
4	5	6	7

delete - 4

5	6	7

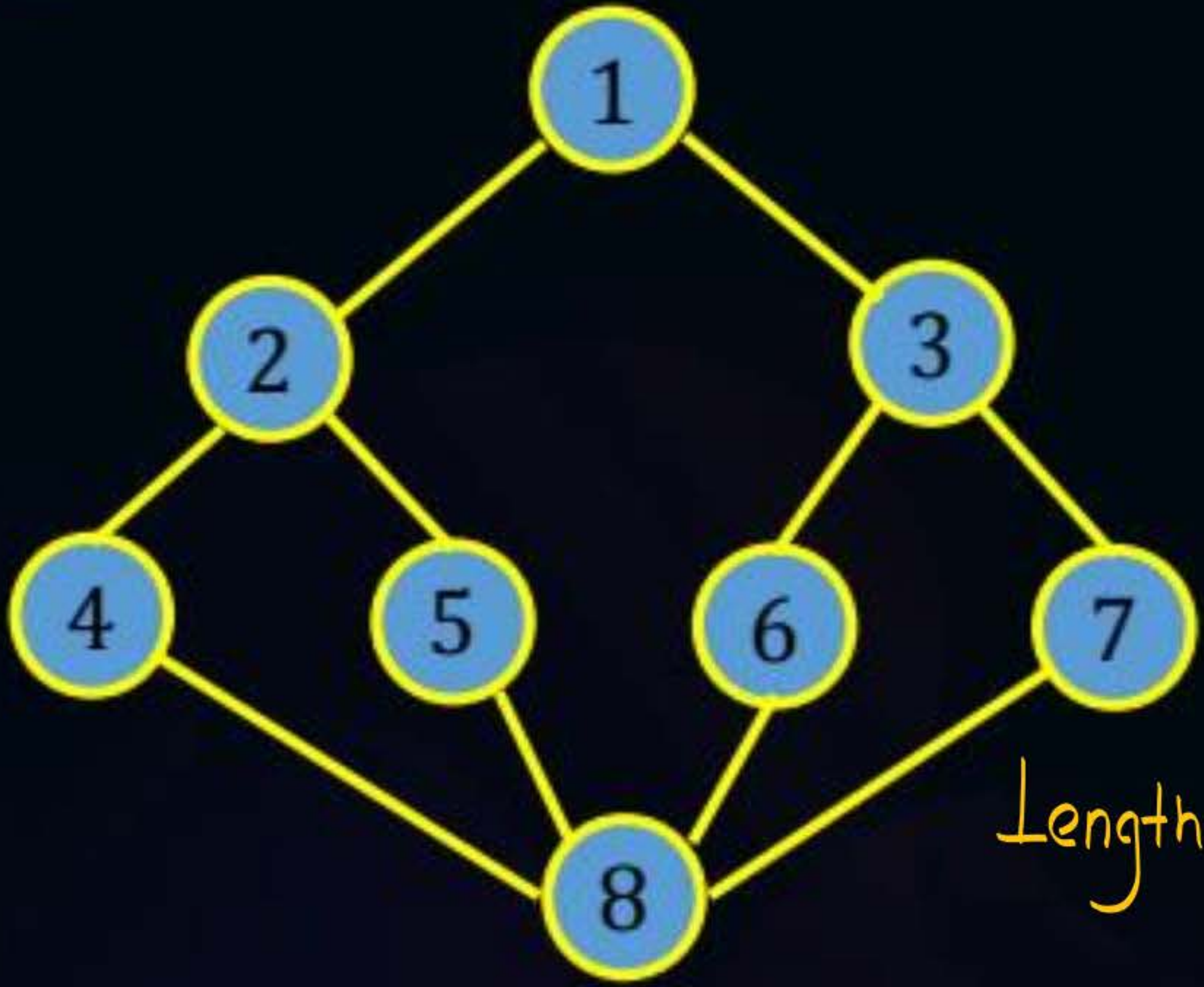
2	3	3	4	4
5	6	7	2	8

⋮

Breadth first search tree

Slide

Topic : Graph

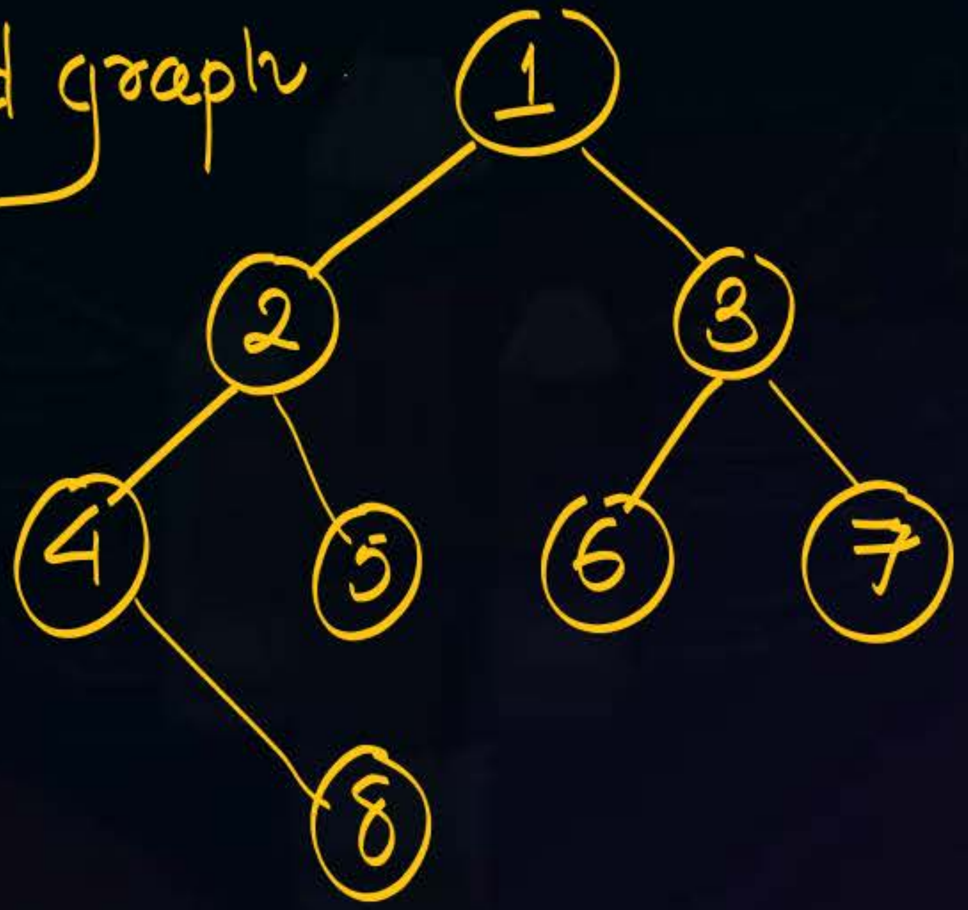


BFS Algorithm

Single source shortest path algorithm

for unweighted graph

Length is No of edges



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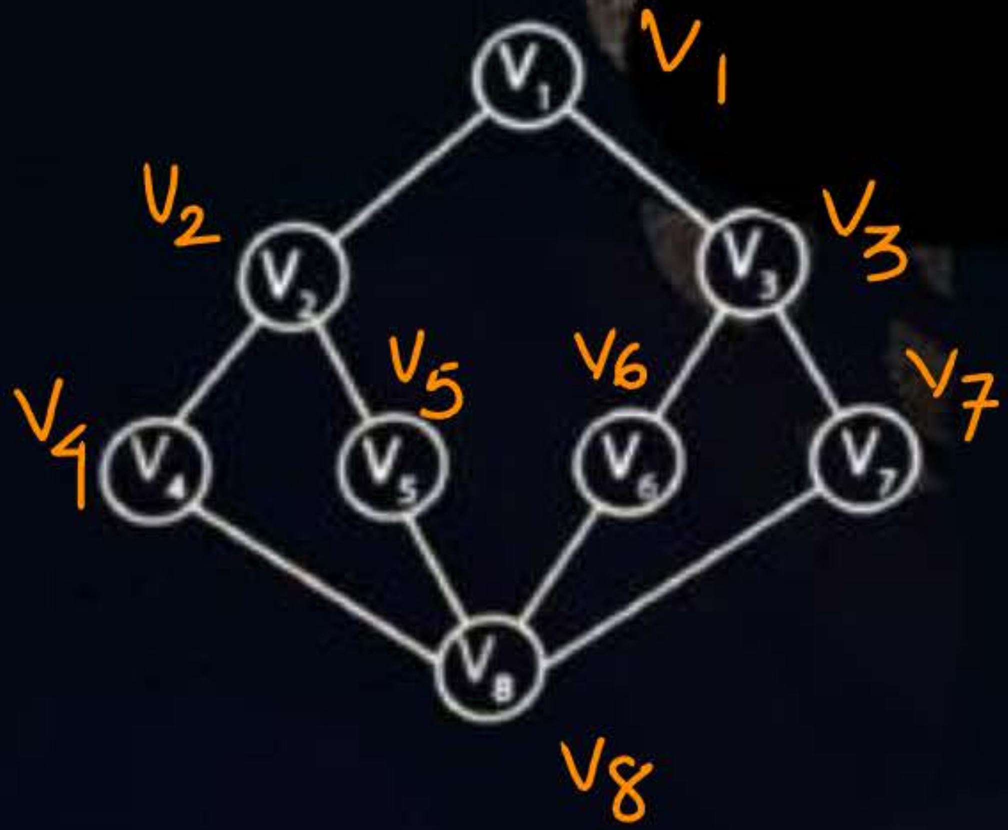
Question

Consider the following graph: [redacted]



1, 3, 2

1, 2, 3

1, 2, 3

1, 3, 2, 7, 6, 5, 4, 8

Identify **valid BFS** search sequence

(a) 1, 3, 2, 5, 4, 7, 6, 8 ✗

(b) 1, 2, 3, 6, 7, 4, 5, 8 ✗

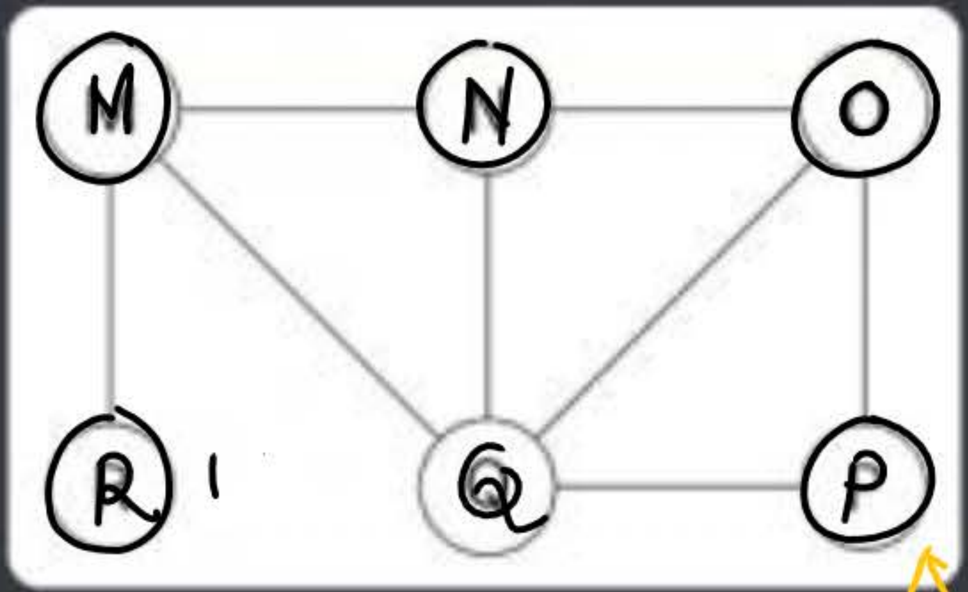
(c) 1, 2, 3, 4, 6, 7, 5, 8

(d) 1, 3, 2, 7, 6, 5, 4, 8 ✓

Slide

Question

The Breadth First Search (BFS) algorithm has been implemented using the queue data structure. Which one of the following is a possible order of visiting the nodes in the graph below?



A. MNOPQR
B. NQMPOR
C. QMNROP
D. POQNMR

MNO X

↑

NQMP ← wrong

C QMNR

↑ wrong

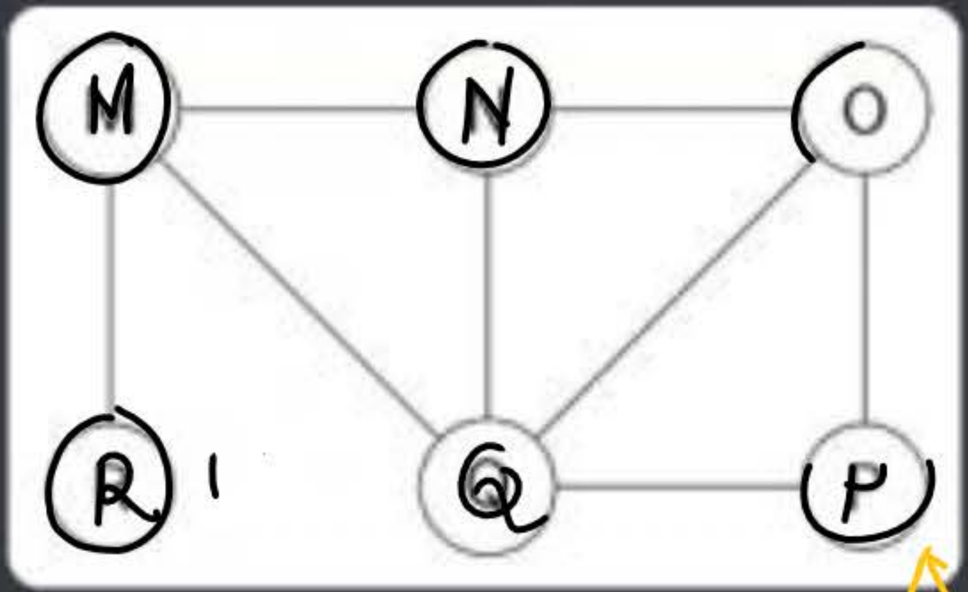
D POQNMR

No of BFS traversal possible considering M as source vertex

Slide

Question

The Breadth First Search (BFS) algorithm has been implemented using the queue data structure. Which one of the following is a possible order of visiting the nodes in the graph below?



M N R Q O P

M N Q R O P

M R N Q O P

M R Q N < P - O / O - P

M Q R N < P - O / O - P

M Q N R < P - O / O - P

No of BFS traversal possible considering M as source vertex

9

Answer

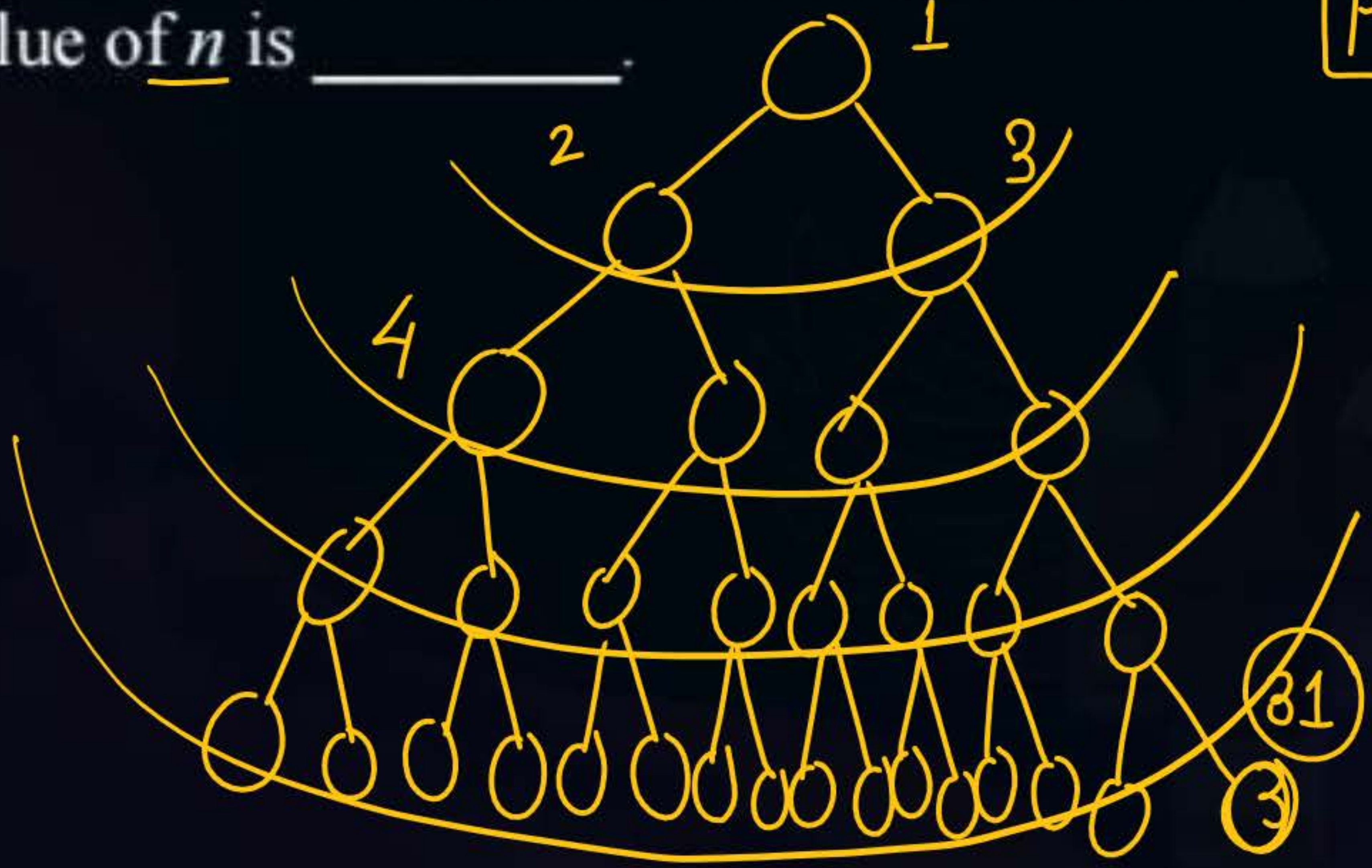
A. MNOPQR
B. NQMPOR
C. QMNROP
D. POQNMR

Slide

Question

Q. Breadth First Search (BFS) is started on a binary tree beginning from the root vertex. There is a vertex t at a distance four from the root. If t is the n^{th} vertex in this BFS traversal, then the maximum possible value of n is ________.

PYQ



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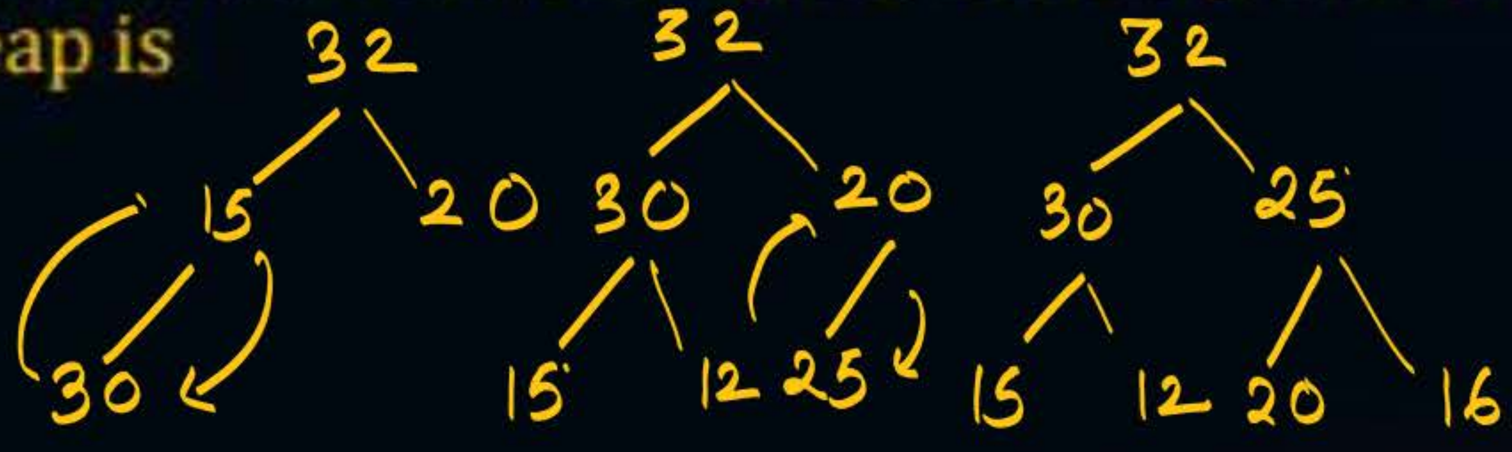
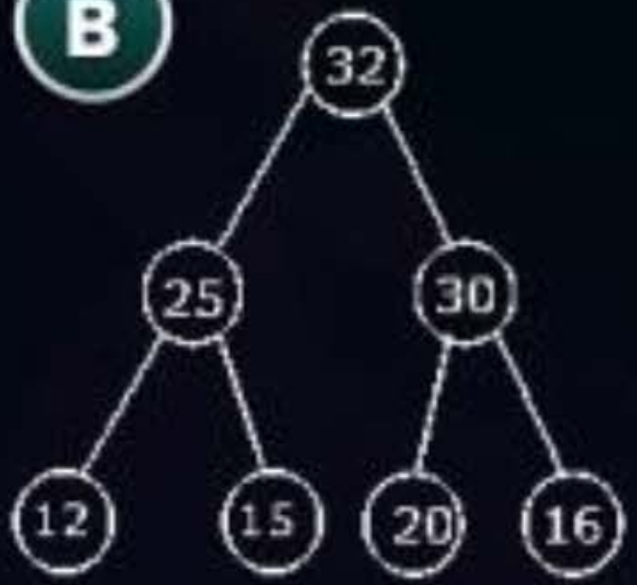
Topic : Heap

#Q. The elements 32, 15, 20, 30, 12, 25, 16, are inserted one by one in the given order into a maxHeap. The resultant maxHeap is

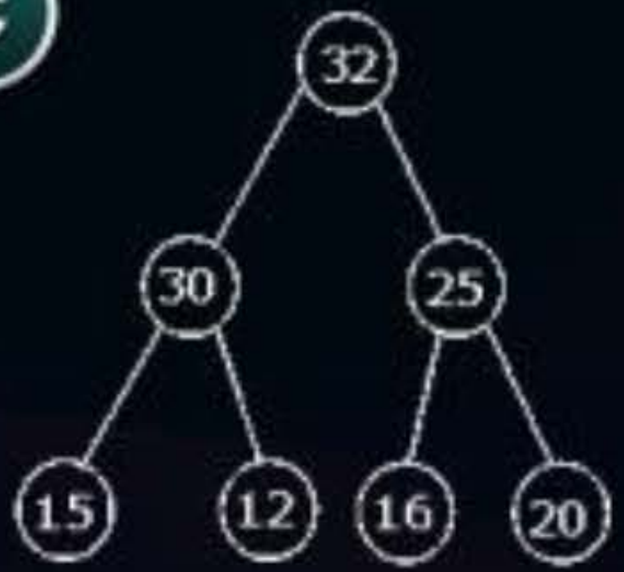


A

B



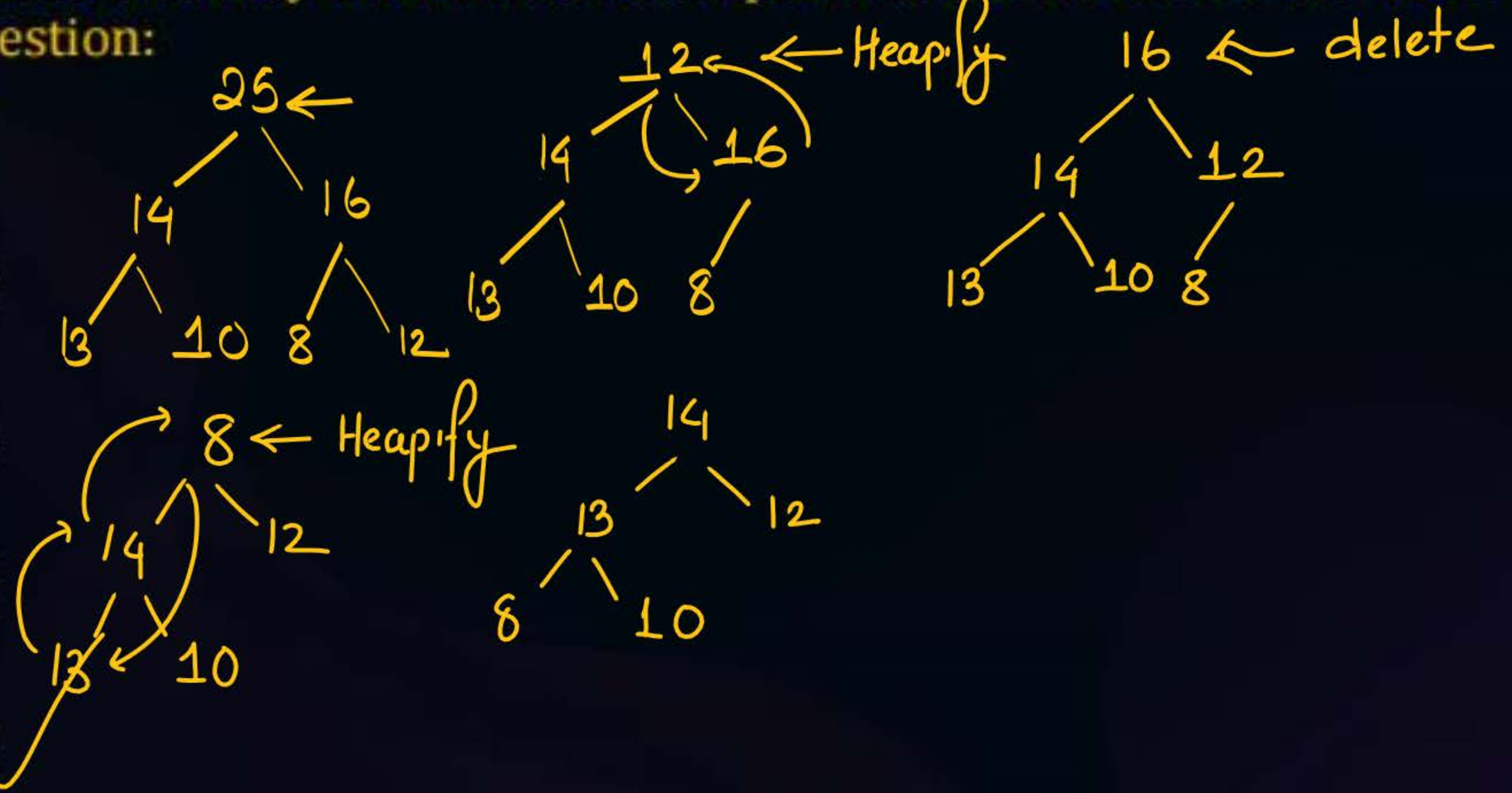
C



D

Question

#Q. If heap is represented by {25,14, 16, 13, 10,8 ,12}
What is the content of the array after two delete operations on the correct answer to the previous question:



(A) {14, 13, 12, 10, 8}

(B) {14, 12, 13, 8, 10}

(C) {14, 13, 8, 12, 10}

(D) {14, 13, 12, 8, 10}

Question

#Q. The minimum number of interchanges needed to convert the array into a max-heap is:
89, 19, 40, 17, 12, 10, 2, 5, 7, 11, 6, 9, 70

(10, 70) ✓

(40, 70) ✓

1. Insert one after another
2. Convert entire array to heap multiple Heapify

[c]

only 2

(A) 0

(B) 1

(C) 2 ✓

(D) 3

Topic : Heap

Consider a binary min-heap containing 169 distinct elements. Number of leaf present in heap is ________

No of leaf Node . $\left\lceil \frac{n}{2} \right\rceil$ CBT

$= \left\lceil \frac{169}{2} \right\rceil = \underline{85}$

No. of internal node

$\left\lfloor \frac{n}{2} \right\rfloor = \left\lfloor \frac{169}{2} \right\rfloor = 84$

Slide

Question: Heapify

An array [82, 101, 90, 11, 111, 75, 33, 131, 44, 93] is heapified. Which one of the following options represents the first three elements in the heapified array?

(A) 82, 90, 101

(B) 82, 11, 93

(C) 131, 11, 93

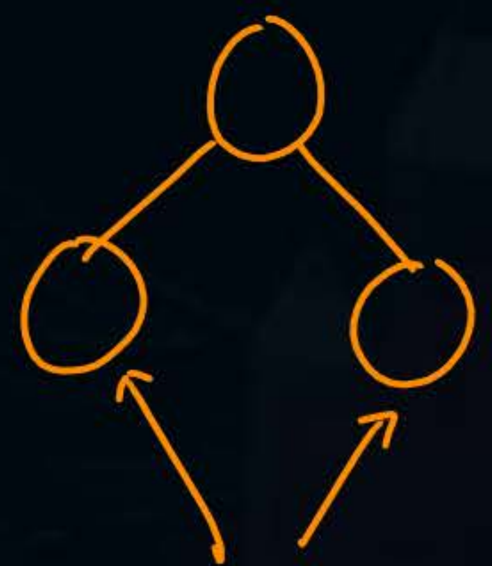
(D) 131, 111, 90

Max heap 131

Min element will always a leaf Node

131 . Max

11 min



Second Smallest element

Slide

Full Length Test

THANK - YOU

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